James Akl

Robotics Researcher

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jamesakl.com | GitHub | ORCiD | Worcester, MA

EDUCATION Coursework

PhD (ABD)	Robotics Engineering	Worcester Polytechnic Institute	GPA 4.00/4.00	**Aug 2019**	**May 2024** (exp.)
MS	Robotics Engineering	Worcester Polytechnic Institute	GPA 4.00/4.00	**Aug 2019**	**May 2021**
BEng	Mechanical Engineering	Lebanese American University	GPA 3.72/4.00	**Sep 2015**	**May 2019**
Minor	Mathematics	Lebanese American University	GPA 4.00/4.00	**Sep 2015**	**May 2018**

EXPERIENCE

Doctoral Researcher — Worcester Polytechnic Institute
Manipulation & Environmental Robotics Laboratory — **Aug 2019 – PRESENT**

Leading a **complex role** requiring **technical proficiency**, **organizational meticulousness**, **tact**, **discipline**, **perseverance**, and **patience**.
Developing **strategies/algorithms/AI** for metal cutting and waste sorting using **robot vision**, **control**, **planning**, and **machine learning**.

Research	Performed research duties as a PhD candidate, working with other researchers and our lab director.
Collaboration	Worked closely with researchers, practitioners, workers, and managers from different fields.
Leadership	Managed and worked with teams of other researchers across diverse projects.
Presenting	Communicated research updates/results to technical and general audiences (company directors).
Engineering	Researched, designed, and implemented innovative solutions to challenging technical problems.
Programming	Developed software for simulations/experiments (C++, Python, ROS, Bash, Linux, Git, PCL, OpenCV, …).
Hardware	Used physical robots, {RGB-D, NIR} cameras, distributed systems, microcontrollers/SBCs, and gas torches.
Experiments	Designed and built real-robot experiments, data collection setups, hot work area and torch cutting setups.
Safety Logistics	Managed safety logistics, certifications, and permits (*e.g.*, NFPA Safety Training, Hot Work Permit, Gear).
Recordkeeping	Maintained an organized environment across several platforms to keep lab operations efficient.
Field Work	Traveled to survey on-site operations in recycling facilities (MA, VT), and in shipbreaking yards (TX).
Publishing	Drafted manuscripts detailing our work; accepted among the top conferences and journals in the field.
Documentation	Wrote work guides for new colleagues to take over my duties and trained them for work procedures.
IP Law	Worked with patent attorney to draft invention disclosures and USPTO patent applications.

Research Mentor — Worcester Polytechnic Institute
Robotics Engineering Department / NSF REM Program — **May 2022 – Jul 2022**

Mentored students in **research methodology**, **data collection/processing**, image processing, robot vision, and **hands-on experiments**.

Teaching Assistant — Worcester Polytechnic Institute
Robotics Engineering Department — **Aug 2019 – May 2020**

Assisted **delivery**, **labs**, and **grading** of robotics courses: *Unified Robotics I: Actuation*, *Unified Robotics II: Sensing*, *Introduction to Robotics*.

Visiting Researcher — University of Ottawa
Mathematical Modeling & Physical Experiments — **Jun 2018 – Aug 2018**

Modeled the vibrations of ribbed plates, and **experimentally verified** the performance of **analytical** versus **numerical models**.

Research Assistant — Lebanese American University
Multiphysics Modeling & Numerical Simulation — **Nov 2017 – Mar 2019**

Demonstrated the viability of the *ionic buoyancy engine* for underwater propulsion using a coupled multiphysics **numerical model**.

SKILLS Projects

Robotics	Kinematics & Dynamics, Control, Motion planning, Manipulation & Grasping, Robot autonomy
Computing	Algorithms, Data structures, Computer vision, Machine learning & AI, Geometric algorithms
Software	C++, Python, Bash, Git, Linux, ROS/Gazebo, MATLAB/Simulink, Mathematica, CMake
Mathematics	Linear algebra, Probability & Statistics, Analysis & Optimization, Numerical algorithms, Proofs
Research	Surveys/reviews, Mathematical modeling, Simulations, Physical experiments, Field work

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SKILLS (Continued) Projects

Publishing	MS Office, LaTeX, Technical writing, Photo-editing, Scientific illustration/diagrams, Video-editing
Social	Presentations, Communication (technical/general audiences), Collaboration (Cross-disciplinary)
Language	English (native fluency), French (professional fluency), Arabic (professional fluency)

RESEARCH Preprints

[2022][1] **Feature-Driven Next View Planning for Robotic Cutting in Metal Scrap Recycling** *[Under Review]*

[2022][4] **ZeroWaste Dataset: Towards Deformable Object Segmentation in Extreme Clutter** ***IEEE CVPR***

[2021][1] **Towards Robotic Metal Scrap Cutting: A Novel Workflow and Pipeline for Cutting Path Generation** ***IEEE CASE***

[2021][2] **ECNNs: Ensemble Learning Methods for Improving Planar Grasp Quality Estimation** ***IEEE ICRA***

[2020][1] **Comparing & Optimizing Analytical, Numerical & Experimental Vibration Models for [...]** ***IStructE/Elsevier Structures***

[2019][4] **Simplified Setup for the Vibration Study of Plates with Simply-supported Boundary Conditions** ***Elsevier MethodsX***

[2019][1] **Ionic Buoyancy Engines: Finite Element Modeling & Experimental Validation** ***SPIE Smart Structures***

[1–4] Superscripts indicate authorship position in co-authored publications.

AWARDS Proof

Graduated with Distinction, 2019
LAU School of Engineering

LAU Most Novel Club Award, 2018
Founded the Emerging Technologies Club

LAU Engineering Scholarship, 2015
Won a design and programming competition

CERTIFICATION

Hot Work Safety Practices, 2022
National Fire Protection Agency